Ledvin Manuel Leiva Mata

Software Engineer | Platform Engineering, DevOps & Infrastructure

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Profile

Platform-oriented software engineer with hands-on experience building and operating production systems across AWS, Kubernetes, Terraform, PostgreSQL, Docker, and CI/CD. Experienced in AI platform infrastructure, MCP integrations, OAuth2/OIDC security, database reliability, observability, internal developer tooling, and backend development with Python and FastAPI.

Experience

TradeStation, Agentic Platform Engineer (Internship) – San José, Costa Rica (Hybrid) June 2026 – present

- Evaluated PostgreSQL high-availability options and helped migrate production platform databases from Kubernetes to AWS RDS Multi-AZ using Terraform, including backup retention, recovery planning, validation, CI workflows, and migration runbooks.
- Implemented AWS Bedrock Guardrails across AI platform services, integrating configuration through Helm and provisioning automation while adding CloudWatch logging to validate protected model invocations.
- Developed and hardened OAuth2/OIDC flows for MCP integrations, working with dynamic client registration, PKCE, redirect URI validation, authorization code protection, Keycloak, and token lifecycle issues.
- Contributed to ContextForge and internal developer platforms through database schema migrations, federated gateway authorization, MCP catalog integrations, and repository workflow improvements.
- Built and maintained automated testing infrastructure with Playwright and reusable SSO authentication, supporting end-to-end validation of internal AI platform workflows.

Delfix, Infrastructure & DevOps Engineer (Full-time) – San José, Costa Rica (Hybrid) Aug 2025 – Jan 2026

- Designed and deployed hybrid on-premise and cloud infrastructures from scratch, covering networking basics, access control, and service orchestration.
- Implemented an end-to-end monitoring stack with Prometheus, Grafana, Alertmanager, and Blackbox, including per-client dashboards and alerting.
- Automated provisioning, backups, and system updates with Bash and Python to standardize operations and reduce repetitive manual work.
- Hardened ingress with Nginx reverse proxy plus SSL/TLS and enforced least-privilege access for servers and services.
- Wrote deployment and disaster-recovery runbooks to improve reproducibility and maintainability.

5CA, Technical Support / Game Support – Netherlands (Remote) Aug 2024 – Feb 2025

- Provided remote technical support for international clients in the gaming industry, assisting users with account, platform, and gameplay-related issues.
- Managed support cases through tools such as Zendesk, Confluence, Teams, Slack, Verint, and GoLance.
- Investigated incidents, documented solutions, and followed internal processes to ensure accurate and efficient case resolution.
- Strengthened skills in troubleshooting, customer communication, documentation, and remote collaboration in international environments.

Projects

Odoo Module Update Automation Platform

Web platform to orchestrate Odoo module updates and operational tasks across registered client servers via SSH with separate production and demo workflows.

- Deployed a React front end and FastAPI back end with Docker on an OVH VPS.
- Implemented SSH-based orchestration to run controlled updates on registered servers with environment isolation.
- Created GitHub-driven workflows to pull targeted changes and apply them consistently.
- Automated database dump creation and on-demand delivery for customers.

Glamping Pod 3D Builder
Interactive 3D configurator where users customize glamping pods by changing materials, textures, and modular components.

- Implemented real-time material and component toggles in Three.js for a smooth customization flow.
- Built an asset pipeline from Revit to Blender for cleanup and optimization before exporting web-ready models.
- Optimized models and scene performance to keep interactions responsive in the browser.

Monitoring Stack: Prometheus + Grafana + Alertmanager + Blackbox
Observability stack for service availability and resource monitoring across environments and clients.

- Defined alert rules and notification flows for availability and operational issues.
- Built dashboards for per-client visibility and faster incident triage.
- Added blackbox checks to monitor endpoints externally and detect regressions early.

SINPE Móvil & Bank Transfer Payment Validator
Automated validator that matches SINPE and bank transfer payments against orders by sender, amount, and date, updating order status when validated.

- Validates payer identity, amount, and timestamp against order data.
- Flags mismatches for manual review and keeps an audit trail of validations and status changes.

Education

Tecnológico de Costa Rica, B.Eng. in Computer Science – Cartago, Costa Rica — Expected Jun 2027

Colegio Científico de Costa Rica, High School Diploma in Science – Pérez Zeledón, Costa Rica — Dec 2022

- Second Place at Feria Científica Nacional de Costa Rica.
- Top 10 admission score for Tecnológico de Costa Rica.
- Astronomy Olympiad Second Place.

Skills

Backend: Python, FastAPI, Flask, REST APIs, Bash/Python automation

Frontend: JavaScript (ES6+), React, basic Three.js

Platform Engineering: Kubernetes, Terraform, Helm, Docker, CI/CD, internal developer platforms, MCP integrations, platform automation

DevOps & Infrastructure: Linux (Ubuntu), Nginx, Git, GitHub Actions, Prometheus, Grafana, Alertmanager, Blackbox Exporter

AWS & Cloud: AWS RDS, Bedrock, CloudWatch, EC2, S3, OVH, DigitalOcean, Contabo, infrastructure as code, high-availability architectures

Security & Identity: OAuth2, OIDC, PKCE, Keycloak, SSO, dynamic client registration, access control

Databases: PostgreSQL, MySQL, MongoDB, Redis

Testing: Playwright, end-to-end testing, SSO-authenticated test workflows

Languages: English (C1), Spanish (Native)